

Sofull is working on improving the package, driver and lamp shell. According to OSRAM, the chip and package technology improvement is very important to increase the efficacy and size of the LED.

According to Everlight, this is mainly a chip technology improvement. The company looks into smaller chips that have greater efficiency and can be overdriven. These types of chips need to be paired with packages that are specifically designed to withstand higher drive currents while maintaining the same volume.

Another area of significance is how makers are handling the process of matching LED chips of different wavelengths with different phosphor formulae to eliminate the different binning processes required for different LEDs. Everlight is using a combination of chip versus phosphor algorithms paired with proprietary ways of depositing the phosphor onto the chip to maintain tighter production output. Sofull will distinguish the LEDs

strictly, based on optical data, different wavelengths and phosphor powder working for different products. OSRAM is steering towards production stage.

Increasing lifespan and other improvements

It is said that capacitors can help in increasing the lifespan of LED lighting products. Normally, the lighting system dies due to the capacitor's lifetime, not the LED's lifetime. According to the Everlight spokesperson, many times when an LED fixture dies, it is not the LED that dies but the components inside the power supply/drive—notably, the capacitor. Cost-effective and more robust capacitors will definitely increase the overall lifetime of the fixtures across all applications. Sofull adopts bile capacitors to replace electrolytic capacitors.

Some improvements are also happening in heat dissipation. OSRAM is improving the package design to decrease the thermal resistance. Also,

higher lm/W can convert more electric energy into light energy instead of heat energy. Everlight is looking for new packaging materials like special plastics or ceramics that allow for better heat dissipation and overdriving of the entire package. As per Sofull, graphite allows better heat dissipation than aluminium.

Smart LED lighting systems are no longer limited to being conceptual prototypes. Sofull has already produced LED smart light, such as microwave sensor LED light.

Finally, how can the LED uptake be increased? As the number of LED chips needed is reduced, overall lighting costs and the power consumption will reduce. This, in turn, will improve LED usage rate. Increasing the LED's efficiency and reducing cost by reaching large volumes is necessary for the increase in adoption of LED in the lighting industry. ●

The author was executive editor at EBY till recently

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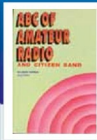
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